EECS Capstone

Day 1

Course No.: EECS Capstone | Credit Hours: 3
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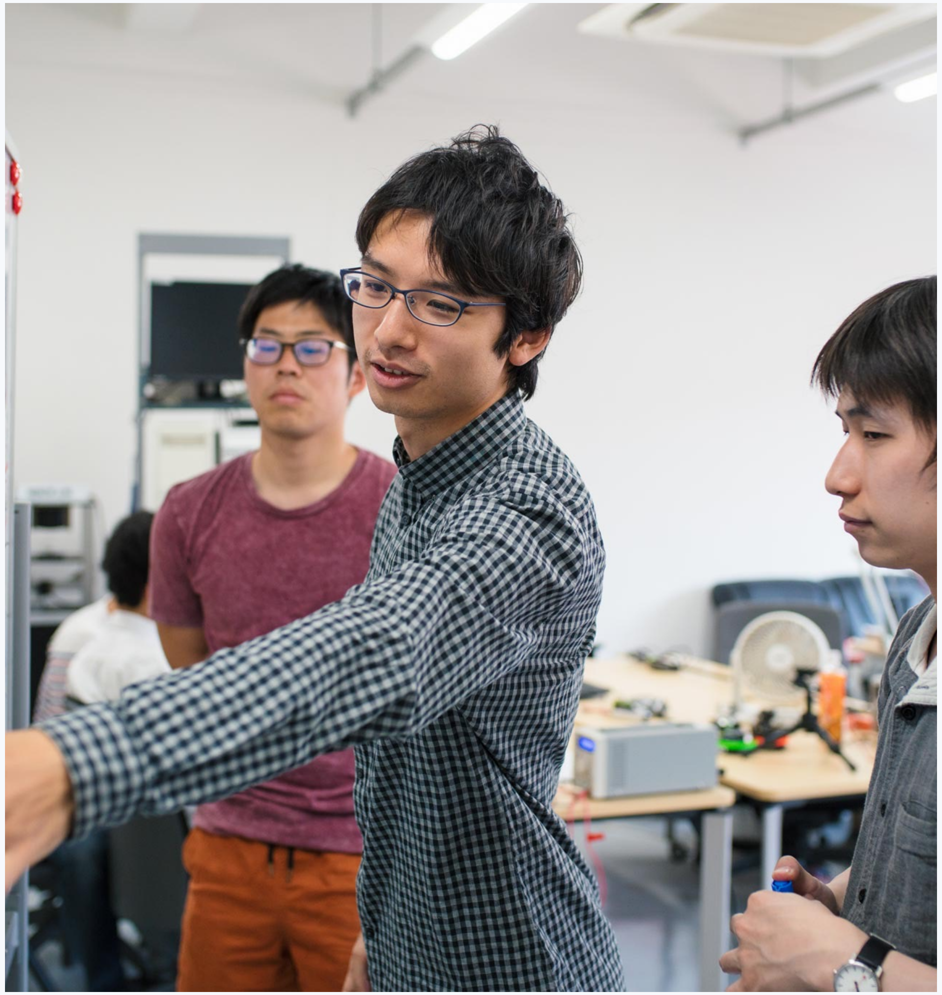
This class simulates real engineering

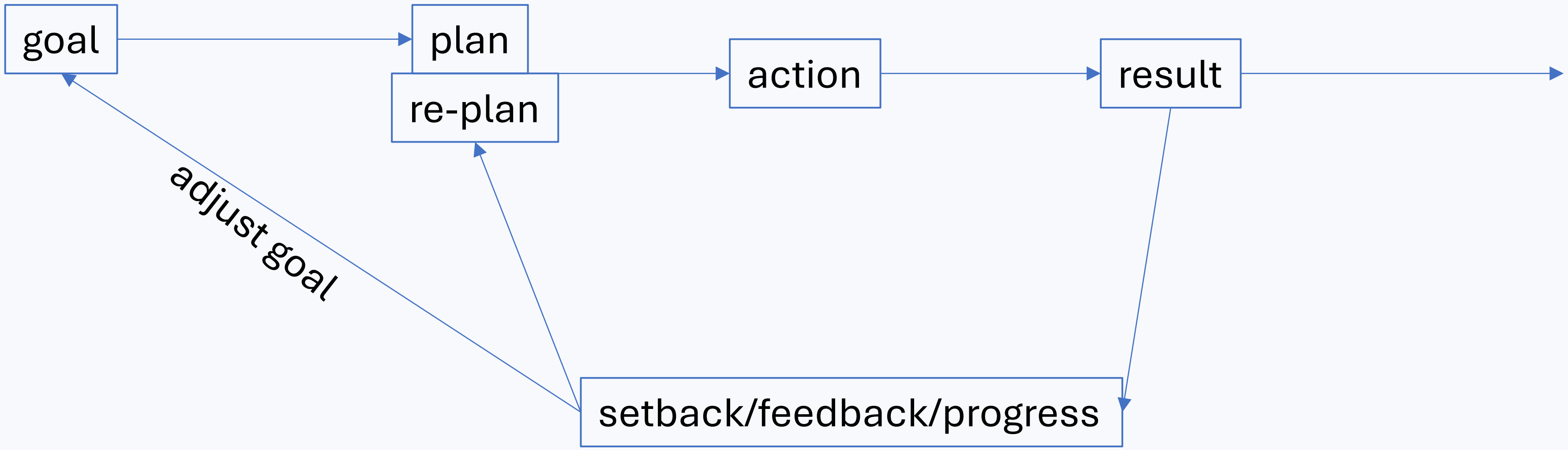
Project-based, team-based, and intentionally imperfect.

REAL-LIFE PRACTICE

Capstone is a class, but it should feel like a small engineering company.

You will face ambiguity before you have answers.
Your first plan will be incomplete.
Progress comes from action, reflection, and updated plans.




goal
plan
re-plan
action
result
adjust goal
setback/feedback/progress

My summer project: reseeding the lawn

A simple yard job turned into a useful capstone lesson.

The HOA asked me to remove weeds. I decided to clear the lawn and reseed it. But I had never done it before, so I kept delaying until the deadline arrived.

Monday: trimmed trees instead.
Tuesday: researched fertilizer; learned local restrictions.
Wednesday: finally started at 3 p.m.

I don’t have a concept of how much i can do, i thought i can get it done in one afternoon.

Plan A: I have an instinct, I should remove the weed first, and dig the dirt, plant the seeds, and water it.

After two hours, only about 10% was done. And that night, I got heat stroke. Had to rest for a week.

What changed after I started?

The task became less frightening after the first imperfect attempt.

1

Start small

I forced myself to work for about 30 minutes instead of waiting for a perfect day.

2

Learn from action

At first I used a hoe. Then I found a rake could open shallow soil faster and safer. Because there are cables underground.

3

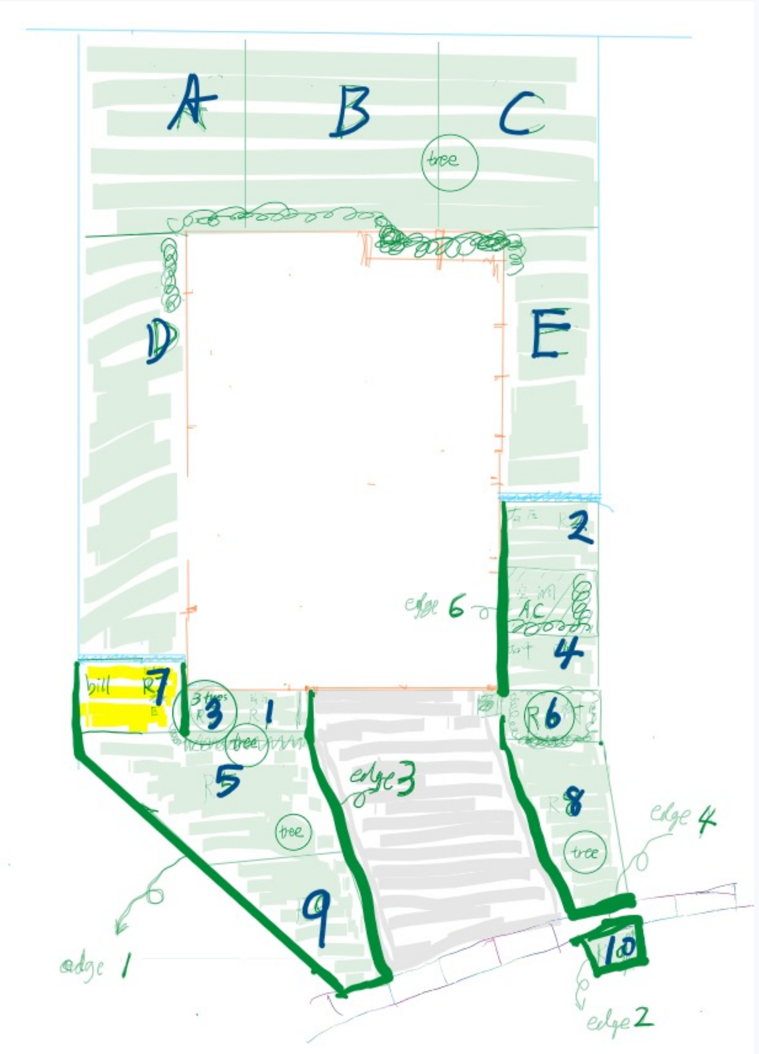
Divide and conquer

I numbered the lawn into 10 areas. 1, One big scary job became 10 smaller targets. 2, Before I do this, I never pay attention to region 2, 3, 7, because they are shared with my neighbors, and they are blocked by bushes.

4

Test the obstacle

I expected the hose connection to need an adapter. mentally, i want to avoid it, so I don’t need to face the expected obstacle. However, I tried it anyway; it worked because the previous house owner left his adapter.



Action created feedback. Feedback created a better plan.

reseeding the lawn: Re-Plan

A simple yard job turned into a useful capstone lesson.

Plan A: I have an instinct, I should remove the weed first, and dig the dirt, plant the seeds, and water it.

After plan A: I learned how to properly seeding. I need fertilize the seed.

Went to Lowe's, the garden guy give me something to read.

FL banned fertilizing between April and September.

This is a constrain, and a setback. I start to do more research about it. It turns out FL soil is nutrition enough for most of the time, it's not necessary to fertilize the grass.

New information comes, re-plan. I can just water it, it should be good.

Two difficulties you will also meet in capstone

1 Fear of unfamiliar work

The brain dislikes unknown tasks because it cannot estimate the cost. You may avoid the project by doing easier, familiar work first.

2 Small setbacks break momentum

Every time you finally start, a small issue appears: missing parts, confusing requirements, unclear roles. It feels like there is no quick positive feedback.

A practical way to handle projects

1. **Start**

   Work for 30 minutes, even if the method is imperfect.

2. **Divide and conquer**

   Break the project by function, space, and time. This is part of your plan and re-plan process.

3. **Re-plan**

   Turn each obstacle into its own smaller task. Treat each setback as a feedback from reality, and do a re-plan loop.

4. **Be patient**

   Trust the plan; you have two semesters. Don’t try to do everything in one day.

5. **Allow feelings**

   Resistance and frustration are normal.

6. **Stay optimistic**

   Some imagined problems disappear once you act.

Capstone rule: no failure, only information for the next plan.

Professional skills you will practice (ABET student outcomes)

Capstone is where technical work meets ABET-style professional outcomes.

Communication

Explain decisions, progress, and problems clearly to teammates, instructors, sponsors, and users.

Teamwork

Work effectively with different people, assign roles, resolve conflict, and meet shared goals.

Self-Directed Learning

Learn unfamiliar tools, standards, hardware, software, and methods when the project requires them.

Project Management

Break down work, plan tasks, track progress, manage deadlines, and revise plans from reality.

Professionalism

Be reliable, communicate early, document decisions, and keep the team moving under pressure.

Adaptability

Treat uncertainty and setbacks as information for the next version of the plan.

Team roles: define ownership, not boundaries

Project Manager / Scrum Lead

Plans tasks, runs meetings, tracks progress, manages deadlines.

Systems & Technical Lead

Keeps the design coherent and coordinates interfaces.

Design / Implementation Lead

Owns building, coding, fabrication, and integration.

Testing & Validation Lead

Defines tests, metrics, verification, and troubleshooting.

Documentation & Communication Lead

Maintains reports, meeting notes, presentations, and sponsor communication.

Everyone still does engineering work. The role tells us who owns each responsibility.

Professionalism: feelings are real, responsibility remains

You can feel bad and act professionally.

Communicate before disappearing.
Respect teammates even when you disagree.
Separate personal frustration from engineering decisions.
Adjust the plan when reality changes.
Ask for help when necessary.
Stay productive whenever you can.

Problems are acceptable.
Silence and avoidance are much harder to manage.

When the team has two ideas

Professional disagreement has a process.

Disagree before the decision.
Commit after the decision.

1. **Before**

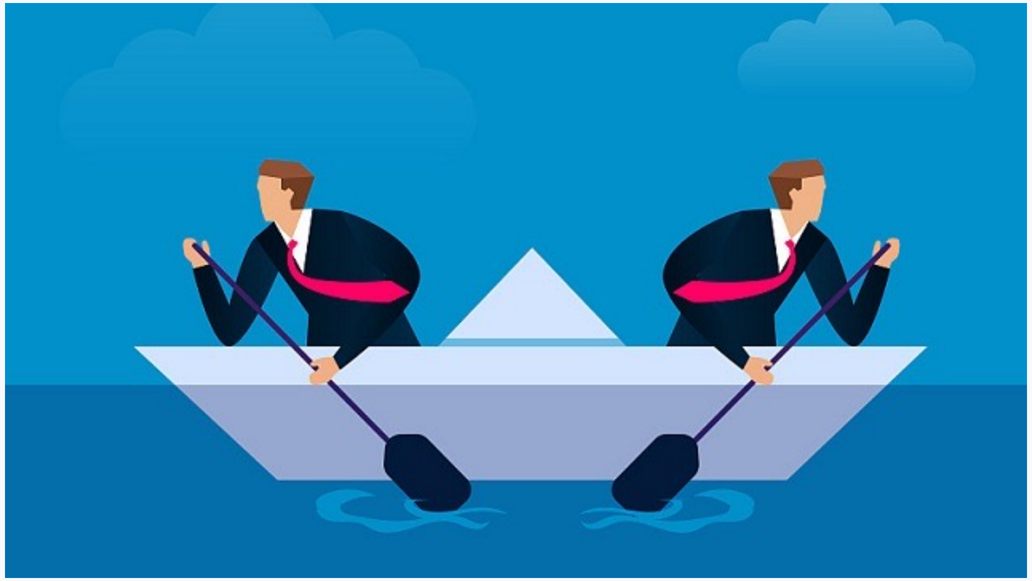
   Bring evidence. Compare requirements, cost, risk, schedule, safety, testability, and sponsor needs.

2. **Decide**

   Make the decision explicitly: data, consensus, vote, technical lead, instructor, or sponsor input.

3. **After**

   Support the selected direction unless new evidence appears. Do not undermine the team.



A professional sentence:

“I still think the other design was better, but this is our team decision. I will do my best to make it work.”

when meet setback: Blame the universe, not your teammate. And re-plan.

Professional does not mean passive

It means disagreement becomes useful information, not team damage.

Do not

Repeatedly reopen the same argument without new evidence.
Do poor work because your idea lost.
Withhold information.
Wait for the project to fail so you can say “I told you so.”

Do

Document why the decision was made.
Bring new test data if the decision should change.
Ask the instructor or sponsor when requirements, safety, or scope are unclear.
Stay useful to the team.

A majority vote is not enough if it violates requirements, safety, or evidence.

Senior Design class Policy

- **Attendance**
  Attendance is required for all class sessions. Instructors and TAs will check attendance.
- **In-Class Progress Checks (weekly)**
  During class, instructors and TAs will visit each group to review progress, answer questions, and provide guidance as needed.
- **Technical and Documentation Feedback**
  TAs may ask detailed technical questions about your project and will provide feedback on project documents throughout the class period.
- **Project Backlog and Sprint Planning**
  Maintain an up-to-date project backlog. At the beginning of each sprint, identify planned tasks and estimate the difficulty or effort required for each task using a consistent method, such as points or estimated hours.
- **Questions, Requests, and Concerns**
  If you have a request, concern, team issue, or other problem, bring it to any instructor or TA. We are happy to discuss sensitive matters privately.

Senior Design Lab Policy

1. **Safety First**
   Follow all lab safety rules, posted procedures, and instructor/TA directions.
2. **Know Before You Energize**
   Check wiring, polarity, voltage/current limits, grounding, moving parts, and other hazards before powering a system.
3. **Work Within Your Training**
   Do not use unfamiliar or hazardous equipment without proper training or approval. If unsure, stop and ask.
4. **Protect People Before Hardware**
   Never take an unnecessary safety risk to protect equipment, save time, or meet a deadline.
5. **Report Problems Immediately**
   Report damaged equipment, accidents, near misses, unsafe conditions, or unusual system behavior to an instructor, TA, or lab manager.

Senior Design Lab Policy

6.Take Responsibility for Equipment
Use tools and equipment correctly, within rated limits, and only for approved project work.
7.Respect Shared Resources
Do not remove, modify, or disturb department equipment, another team's setup, or project materials without permission.
8.Maintain Lab Security
Secure valuable equipment and project materials. Do not share keys, access cards, passwords, or allow unauthorized access.
9.Keep the Lab Professional
Keep work areas clean and organized. Power down equipment, return tools, secure projects, and clean your area before leaving. Don't leave charger on unsupervised.(true story from last semester)
10.Everyone Owns Safety
Every team member has the responsibility to identify hazards, communicate concerns, and stop unsafe work.

Senior Design Materials Policy

Built items, prototypes, purchased parts, tools, and project hardware must stay with the department or customer.

All materials built, purchased, or modified during Senior Design belong to the EECS Department of ERAU or the project customer/sponsor.

Before You Leave

Return project items to the lab, instructor, TA, sponsor, or assigned storage location.

Do Not Take

Do not keep prototypes, parts, boards, sensors, tools, 3D prints, or sponsor-provided materials.

Document It

Label items clearly and tell the instructor/TA where materials are stored.

Professional rule: finish the project, clean up the project, and leave the work for the next team.

When in doubt, ask the instructor before removing anything from the lab.

First week plan

This week

Introduce projects
Understand expectations
Start thinking about teams and roles

next class

Review project options
Ask questions early
Prepare to choose a direction

Welcome to capstone.